

MedVision AI Knowledge Base

Normal Chest Anatomy

1. Overview

Understanding normal chest anatomy is essential for interpreting chest X-rays. Accurate recognition of normal structures helps distinguish normal findings from pathology and reduces false-positive interpretations.

2. Thoracic Skeleton

The thoracic cage consists of the ribs, clavicles, sternum, thoracic vertebrae, and scapulae. These structures protect the heart and lungs and serve as important landmarks on chest radiographs.

3. Airways

The trachea extends from the neck into the thorax and divides into the right and left main bronchi at the carina. A normal trachea is midline or only minimally deviated.

4. Lungs

The lungs occupy most of the thoracic cavity. The right lung has three lobes, while the left lung has two lobes because of the position of the heart. Lung fields should appear well expanded with symmetric vascular markings.

5. Pleura

The visceral pleura covers the lungs, while the parietal pleura lines the chest wall. Normally the pleural space contains only a small amount of lubricating fluid and is not visible on a chest X-ray.

6. Heart and Mediastinum

The mediastinum contains the heart, great vessels, trachea, esophagus, and lymph nodes. On a standard PA chest radiograph, the cardiac silhouette should occupy less than approximately half of the thoracic width in most adults.

7. Diaphragm

The diaphragm separates the thoracic and abdominal cavities. Normally the right hemidiaphragm sits slightly higher than the left due to the liver. Costophrenic angles should be sharp and well defined.

8. Hila and Pulmonary Vessels

The hila contain the pulmonary arteries, veins, bronchi, and lymphatic structures. Pulmonary vessels should taper gradually toward the lung periphery without focal enlargement.

9. Chest X-ray Landmarks

Important landmarks include the trachea, carina, aortic arch, hilar regions, cardiac borders, diaphragms, costophrenic angles, gastric bubble, clavicles, ribs, and thoracic spine.

10. Normal Chest X-ray Appearance

A normal chest X-ray demonstrates clear lung fields, normal heart size, sharp diaphragms, visible costophrenic angles, symmetric lungs, no pleural fluid, no pneumothorax, and no focal consolidation.

11. ABCDE Interpretation Method

A: Airway (trachea and bronchi). B: Breathing (lungs and pleura). C: Circulation (heart and mediastinum). D: Diaphragm (hemidiaphragms and costophrenic angles). E: Everything Else (bones, soft tissues, devices).

12. Common Normal Variants

Common normal variants include an azygos lobe, mildly elevated hemidiaphragm, pectus deformity, prominent epicardial fat pad, and slight tracheal deviation due to the aortic arch.

13. Patient Explanation

A normal chest X-ray means the lungs, heart, and surrounding structures appear healthy without evidence of major infection, fluid, collapse, or other obvious abnormalities. Clinical symptoms may still require additional evaluation if they persist.

14. AI Interpretation

When AI reports a normal examination, clinicians should still correlate the result with symptoms and history. Early disease may occasionally be radiographically occult, and additional imaging or follow-up may be appropriate.

15. Quick Facts

Right Lung: 3 lobes
Left Lung: 2 lobes
Main Airway: Trachea
Pleural Space: Normally not visible
Right Hemidiaphragm: Usually higher than left
Normal Costophrenic Angles: Sharp

16. References

Gray's Anatomy; Netter's Atlas of Human Anatomy; Radiology Masterclass; Radiopaedia educational resources; Merck Manual Professional Edition; standard chest radiology textbooks.